JUNIOR ROBOTICS

MANUAL

Soccer Bot

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AEROBOTICS GLOBAL
TECHNOLOGY THAT EXCITES

CONTENTS

WARNING: The kit contents are not suitable for children under 3 years. Choking hazard – small parts may be swallowed or inhaled.

Store the kit materials and assembled models out of reach of small children.

HANDLING AND PRECAUTIONS

General:

- Assembly and operation of this kit should be done only under the supervision of an adult
- All parts of this kit should be stored and operated away from water and fire sources
- All parts of this kit should never be put into the mouth
- Throwing or dropping the parts may cause damage to the parts

Building/Operating:

- All the tools for assembling must be used with extreme precautior
- The parts should not be forcibly bent, attached or removed
- The parts should not be touched when in motion
- The power supply, motors, sensors and other components should before operating
- The power supply should be turned off when connecting the Arduino to the computer for programming

Batteries:

- The batteries should not be inserted in the battery holder before mounting and connecting the battery holder onto the model
- The batteries should be inserted correctly with the (+) and (-) in the correct direction
- Always use the provided batteries or batteries of correct size and rating (3.7V)
- New and used batteries should not be mixed for operation
- Always close the battery holder with the lid after removal or insertion of batteries

Workshop 1

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Hi Andy, welcome to the next part of our course which is to make a SoccerBot

First, we start revising what we learned in Level 2 course

Do you remember what you learned in Level 2 ?

Okay, I remember we learned how to make a Path Finding Robot and a Social Distancing Robot

Great!!! Let’s revise then with another basic LED program, we will play with the brightness of LED light

Interesting, I’m ready to start

Let’s write a code for increasing the LED brightness from dark to bright

To control the brightness of the LED we will be using PWM signal, this signal can go from 0 to 255, we can assign this to the longer pin(positive) of the LED so that it’s brightness can be controlled

```
int Led_brightness = 0;
void setup() {
  // put your setup code here, to run once:
 pinMode(5,OUTPUT);
 pinMode(6,OUTPUT);
}

void loop() {
  // put your main code here, to run repeatedly:
  digitalWrite(5,LOW);
  for(int i=0; i<256; i++)
  {
    Led_brightness = i;
    analogWrite(6, Led_brightness);
    delay(1000);
  }
}
```

Figure 1.01

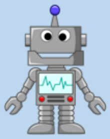
Challenge 1

Decrease the brightness of the LED from bright to dark

Workshop 2

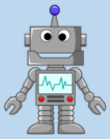
Hi Aebo, what are we doing today?

Today, we are learning how joystick works



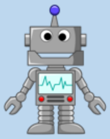
You mean video game Joystick used in PlayStation or Xbox consoles

Yes that’s right



Cool, then I can play even better then!

Let’s start



The Analog Joystick is similar to two potentiometers connected together, one for the vertical movement (Y-axis) and other for the horizontal movement (X-axis). The joystick also comes with a Select switch, when the gear is pressed the switch changes its position and when released, it will go back to alternative position, the joystick has 5 pins as shown in the figure

VCC - Power supply

GND – Ground

VRx – Horizontal control

VRy – Vertical control

SW - Switch

Figure 1.02

The four positions of the gear can be used to control the Forward, Backward, Left and Right directions of the Aebo. Initially when the gear is in idle position the X and Y values are 512,512 which means neutral position and when you move the Joystick gear completely forward, the y axis value goes towards 1023(max value) and if pushed in opposite direction its value goes towards zero, similarly for X-axis, complete Left will go to 1023 and if pushed to complete right will go towards zero

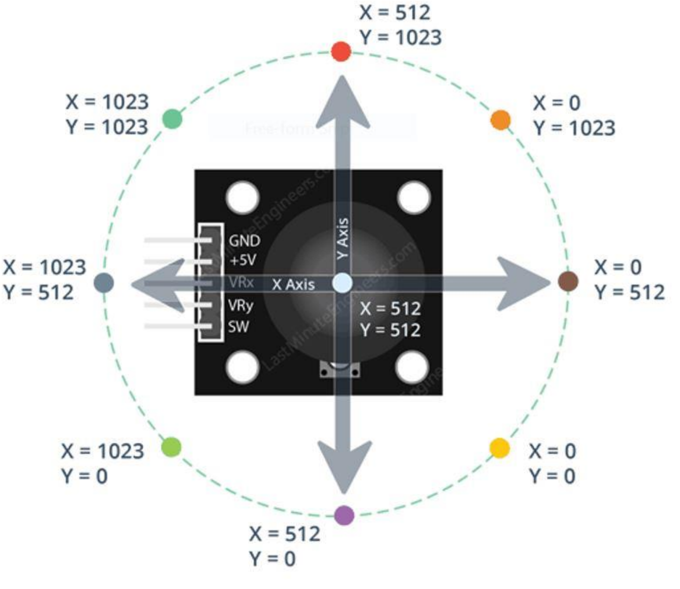


Figure 1.03

W will connect the Joystick to the Robot as shown in the figures 1.04 And 1.05

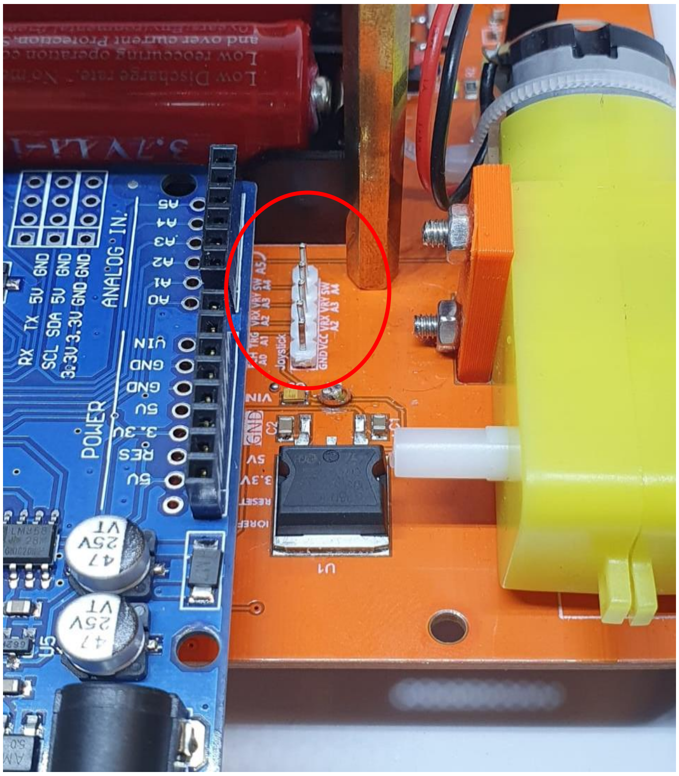


Figure 1.04

Figure 1.05

Let’s write a code for displaying VRx and VRy values on our LCD display to understand how the joystick works

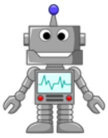
```
#include <LiquidCrystal_I2C.h>

int Vrx = 0;
int Vry = 0;

LiquidCrystal_I2C lcd(0x27,16,2);

void setup() {
  // put your setup code here, to run once:
lcd.begin();
lcd.backlight();
pinMode(A2,INPUT);
pinMode(A3,INPUT);
}

void loop() {
  // put your main code here, to run repeatedly:
Vrx = analogRead(A2);
Vry = analogRead(A3);
lcd.setCursor(0,0);
lcd.print("Vrx value is");
lcd.print(Vrx);
lcd.setCursor(0,1);
lcd.print("Vry value is");
lcd.print(Vry);
delay(100);
lcd.clear();
}
```

Figure 1.06

Workshop 3

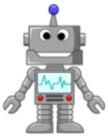
Hi Aebo, what are we doing today?

We are learning how to control our Aebo with the Joystick



That means I can move the Aebo whichever direction I want to!

Yes, you can once you finish this workshop



Can’t wait to play with Aebo, let’s start

We can use earlier defined blocks of Forward, Backward, Left, Right and Stop_robot in our program to control the Aebo using the Joystick

```
int Vrx = 0;
int Vry = 0;
void Left();
void Right();
void Forward();
void Backward();
void Stop_robot();

void setup() {
  // put your setup code here, to run once:
pinMode(A2,INPUT);
pinMode(A3,INPUT);
pinMode(3,OUTPUT);
pinMode(11,OUTPUT);
pinMode(12,OUTPUT);
pinMode(13,OUTPUT);
}

void loop() {
  // put your main code here, to run repeatedly:
Vrx = analogRead(A2);
Vry = analogRead(A3);
if (Vry<200)
{
  Forward();
  }
  if (Vry>800)
{
  Backward();
  }
  if (Vrx<200)
{
  Left();
  }
  if (Vrx>800)
{
  Right();
  }
if (((Vrx>200) && (Vrx<800)) && ((Vry>200) && (Vry<800)))
{
  Stop_robot();
  }

}
```

```
void Right ()
{
       analogWrite(3,100);
       analogWrite(11,100);
       digitalWrite(12,LOW);
       digitalWrite(13,HIGH);
}

void Forward (){
       digitalWrite(12,HIGH);
       digitalWrite(13,HIGH);
       analogWrite(3,100);
       analogWrite(11,100);

}
void Left (){
       analogWrite(3,100);
       analogWrite(11,100);
       digitalWrite(12,HIGH);
       digitalWrite(13,LOW);

       }

       void Backward (){
       digitalWrite(12,LOW);
       digitalWrite(13,LOW);
       analogWrite(3,100);
       analogWrite(11,100);

       }
void Stop_robot (){
       analogWrite(3,0);
       analogWrite(11,0);

       }
```

Figure 1.07

Workshop 4

Hi Aebo, what are we doing today?

We are learning about the Bluetooth module and how it communicates

What is Bluetooth?

Bluetooth is a wireless communication device used for transmission and reception of data

Sounds good, Let's start

Bluetooth Module

The Bluetooth module is designed for serial communication which Transmits and Receives data in a pipeline within a range of 30ft or 10 meters

We use two modules for our Aebo communication, one is connected to Joystick for transmitting commands and the other is connected to the Aebo board for Receiving commands

Before communication, both the Bluetooth modules need to be paired, which is already done before hand

Now when both the Bluetooth modules were turned on, the red light will blink continuously which means it is searching for its pair, if we bring both of them within range then they will communicate and get paired, after that both Bluetooth lights will blink at heartbeat rate

Figure 1.13

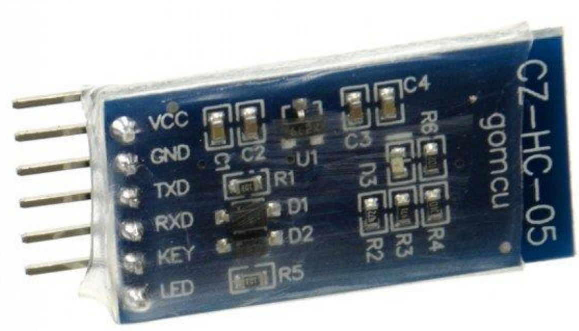


Figure 1.14

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Connect the two joysticks to the Aebo controller as shown in figures 1.16 and insert the Bluetooth transmitter which has T written on its back side as shown in figure 1.17

Figure 1.15

Figure 1.16

Figure 1.17

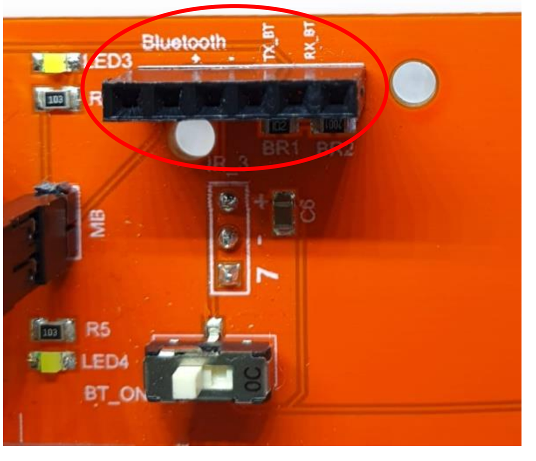


Figure 1.18

Insert the Bluetooth receiver module which has R written on its back side into the robot as shown in figures 1.18 and 1.20

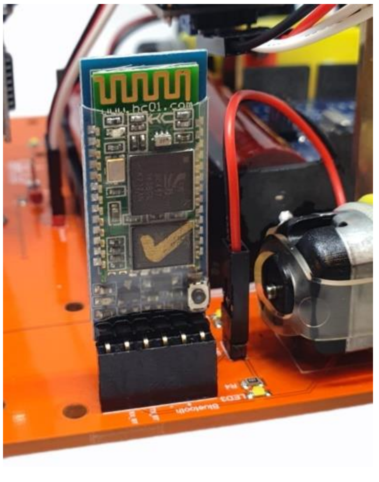


Figure 1.19

Figure 1.20

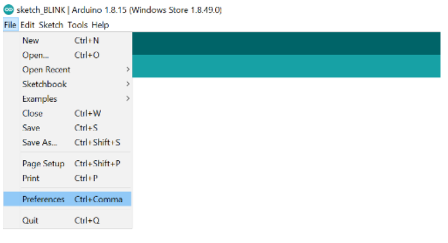
For each time the Joystick is moved from one direction to another, forward , backward, left, or right, the Vrx, Vry and switch values will change and update continuously which makes it difficult to send values through a wireless connection.

Due to this reason we will send a simple character instruction of ‘F’ for forward and ‘S’ for stop and similarly for all other directions. To program the joystick module, we need to follow the following steps.

Step 1 :

- Open Arduino IDE and got to File -> Preferences and paste the following into the Additional Board manager URLs: box as shown in the following figures 1.21.1 , 1.21.2 and 1.21.3.

https://github.com/AeroboticsGlobalTeam/BoardVariants/raw/main/package_AEBO_boards_index.json



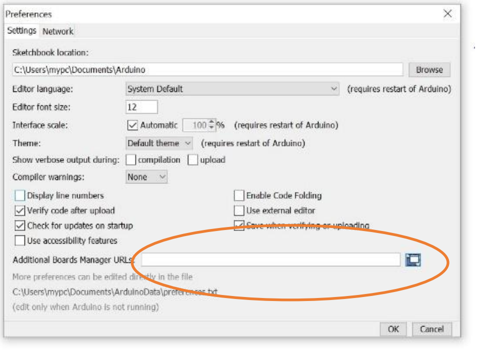


Figure 1.21.1

Figure 1.21.2

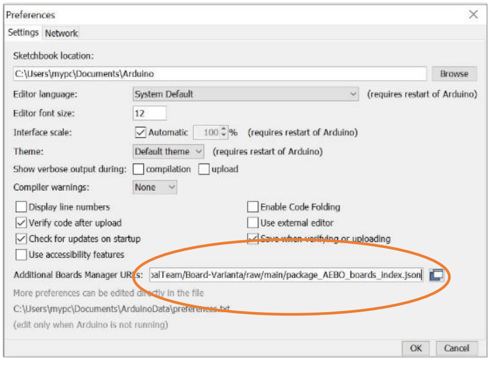


Figure 1.21.3

Step 2 :

Now navigate to Tools -> Board -> Board Manager and install the following boards

- Arduino SAMD boards
- AEBO Joystick

Please use the following figures 1.21.4, 1.21.5 and 1.21.6 for reference.

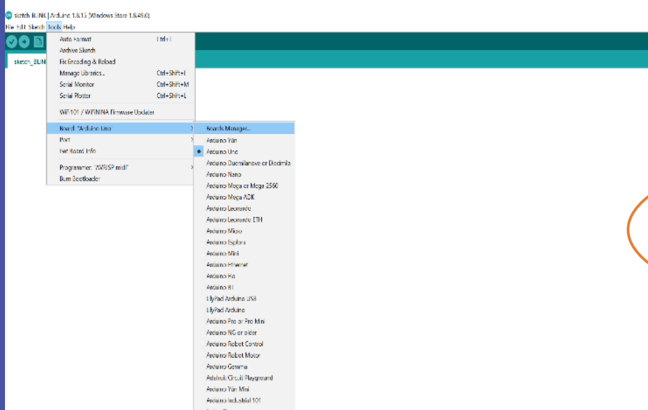


Figure 1.21.4

Figure 1.21.5

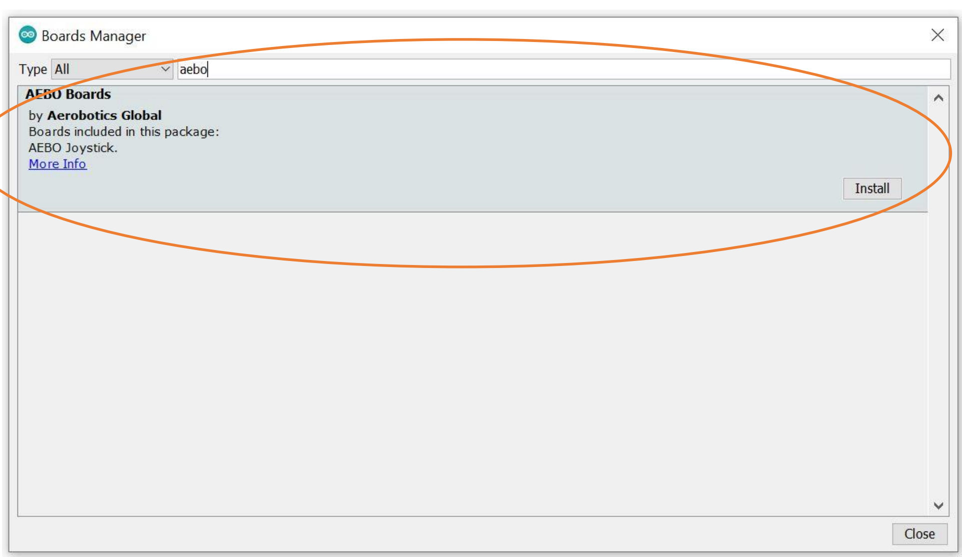


Figure 1.21.6

Now we are ready to program and upload to our joystick

You can use the code from the fig 1.21.7 and upload it to the board by selecting the AEBO joystick board as shown in fig 1.21.8.

```
void setup() {
  // put your setup code here, to run once:

  //Joystick2
  pinMode(8,INPUT); //Xaxis
  pinMode(9,INPUT); //Y axis

  Serial1.begin(115200);

}

void loop() {
  // put your main code here, to run repeatedly:

  VRX2 = analogRead(8);
  VRY2 = analogRead(9);

  if (VRX2 <200){
    Serial1.println('L');
  }

  if (VRX2 >700){
    Serial1.println('R');
  }

  if (VRY2 <200){
    Serial1.println('B');
  }

  if (VRY2 >700){
    Serial1.println('F');
  }

  if(VRX2>200&&VRX2<700&&VRY2>200&&VRY2<700){
    Serial1.println('S');
  }



}
```

Figure 1.21.7

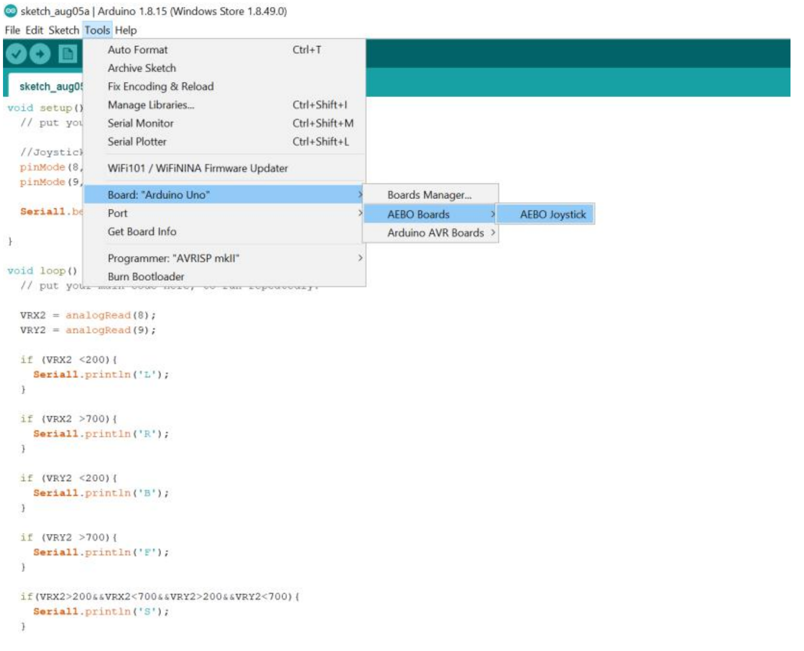


Figure 1.21.8

Workshop 5

Hi Aebo, what are we doing today?

We are adding servo motors to our robot

Okay what is the first step

Take out the servo motors from your kit and be ready with Allen key, screws and nuts

Yes I am ready to assemble, Let’s start

Step 1

First, take two servo motors and insert them into the servo ports with wire going in first and fix them as shown in the figures 1.08 and 1.09

Figure 1.08

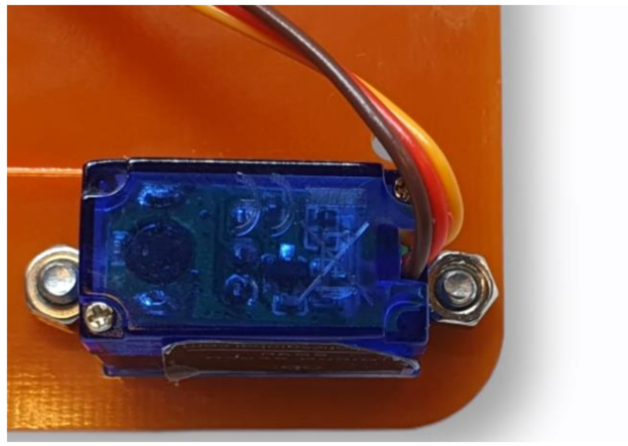


Figure 1.09

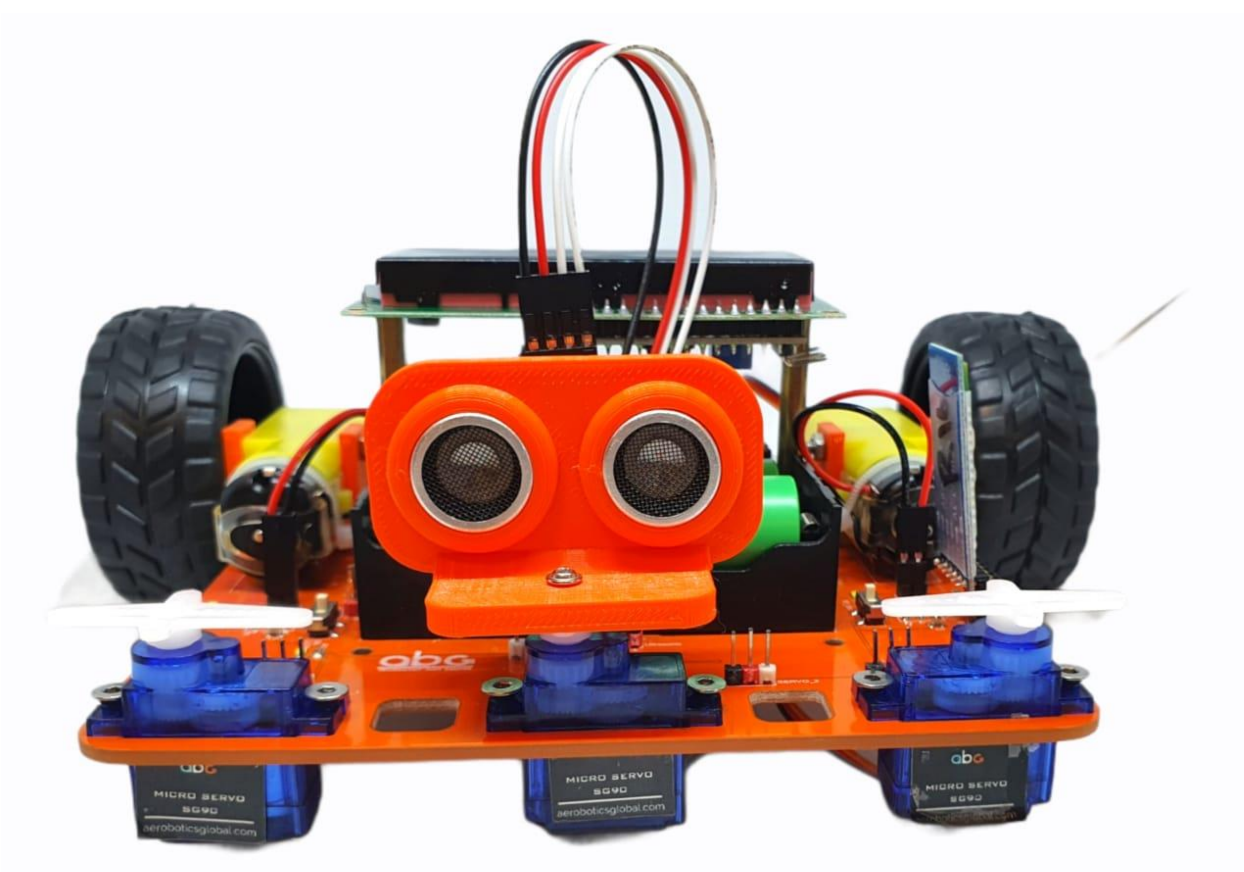


Figure 1.10

Step 2

Connect the servo motors to the P9 and P10 pins with brown wire going to negative as shown in figures 1.11 and 1.12

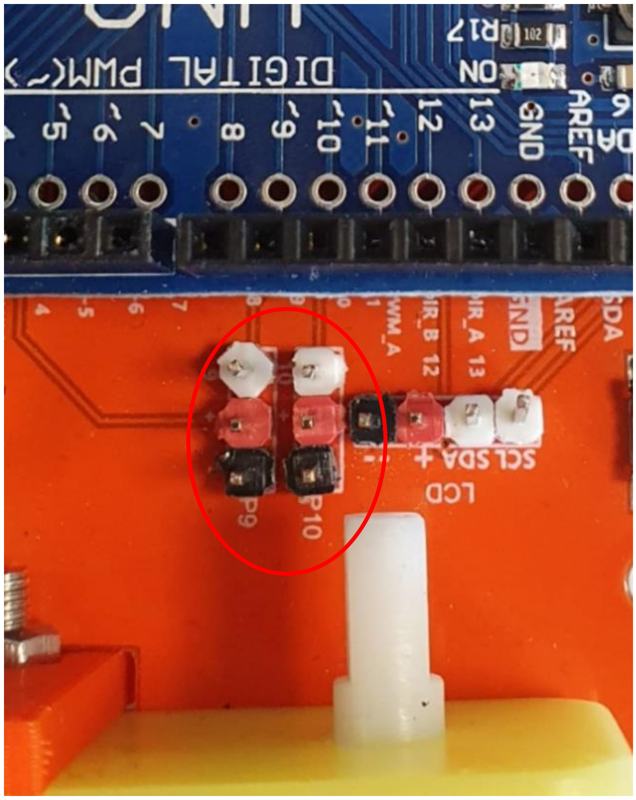


Figure 1.11

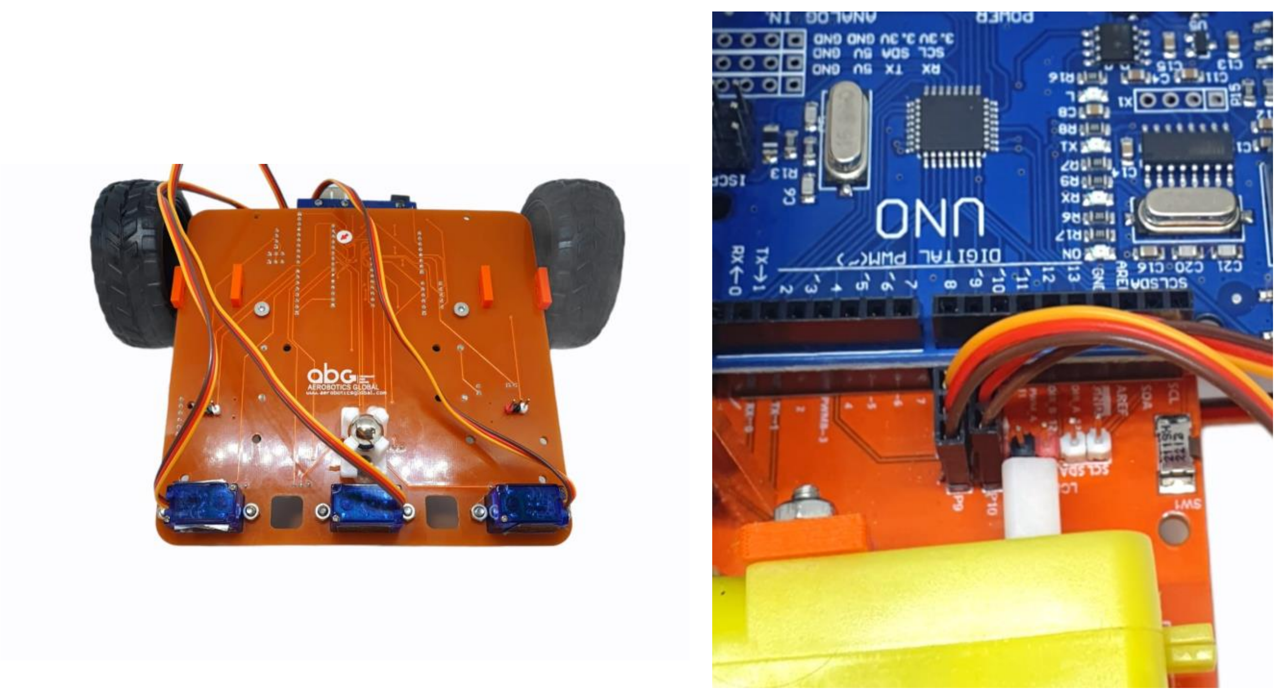
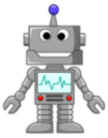


Figure 1.12

Workshop 6

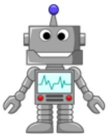
Hi Aebo, what are we doing today?

We are adding flaps to our Aebo, and connecting them to the servo motors



I can see we are about to finish the soccerbot

Yes we are, let's start the workshop then



Step 1

Insert the servo horns into the flaps and place them on top of the servo motors as shown in figure 1.22

Before connecting them make sure left and right flaps are placed correctly and fit them Properly as shown in figures 1.23 and 1.24

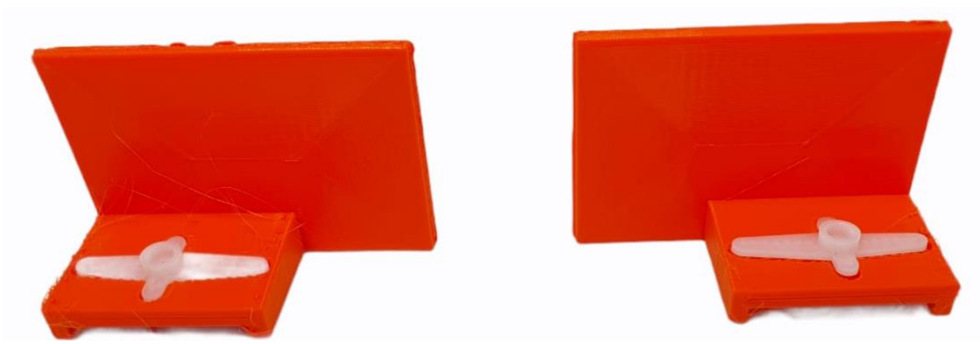


Figure 1.22

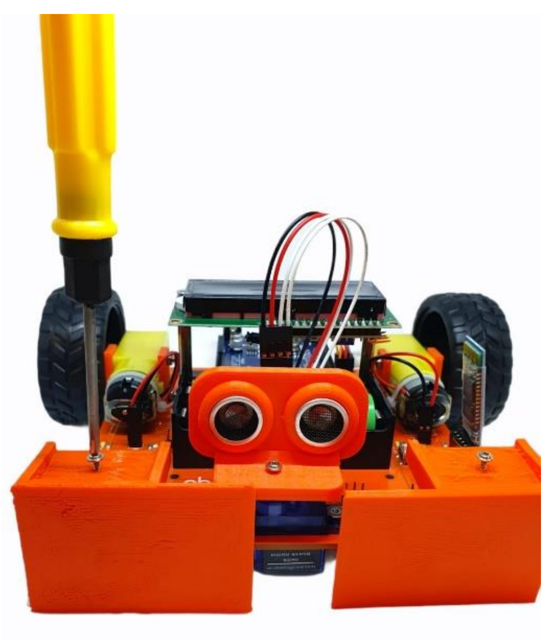


Figure 1.23

Figure 1.24

Now for controlling the flaps, we use the switch button of the joystick,
When the button is pressed both the flaps will be opened and when
released they will close

For Bluetooth we send 'O' to open the flaps with 90 degrees and 'C' to close the flaps
Now make sure for closing the flaps, the one connected on the Bluetooth side is set to 180 degrees because the servo horn is designed to the shorter side and for the other side flap set it to zero degrees
Add the code shown in figure 1.25 to control the flops

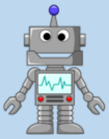
```
void Flaps_open()

{
 servo_9.write(180);
 servo_10.write(0);
}

void Flaps_closed()
{
servo_9.write(90);
servo_10.write(90);
}
```

Figure 1.25

Workshop 7

Hi Aebo, what are we doing today?

We are writing the final code to finish the soccerbot program



Okay! Let's wrap it up then

Upload the final code for **soccerbot** as shown in figure 1.26 and 1.27

```
#include <SoftwareSerial.h>
#include <Servo.h>
int bluetooth;
Servo servo_9;
Servo servo_10;
void Left();
void Right();
void Forward();
void Backward();
void Stop_robot();
void Flaps_open();
void Flaps_closed();
void setup() {
  // put your setup code here, to run once:
pinMode(3,OUTPUT);
  pinMode(11,OUTPUT);
  pinMode(12,OUTPUT);
  pinMode(13,OUTPUT);
  Serial.begin(115200);
  servo_9.attach(9);
  servo_10.attach(10);
}

void loop() {
  // put your main code here, to run repeatedly:
    if(Serial.available() > 0){
          bluetooth = Serial.read();
          if(String(char(bluetooth)) == "F"){
              Forward();

          }
          if(String(char(bluetooth)) == "B"){
              Backward();

          }
          if(String(char(bluetooth)) == "L"){
              Left();

          }
          if(String(char(bluetooth)) == "R"){
              Right();
          if(String(char(bluetooth)) == "S"){
              Stop_robot();

          }
          if(String(char(bluetooth)) == "O"){

Flaps_open();
          }
          if(String(char(bluetooth)) == "C"){

Flaps_closed();
          }
}
}
void Right ()
{
      analogWrite(3,100);
      analogWrite(11,100);
      digitalWrite(12,LOW);
      digitalWrite(13,HIGH);
}

void Forward (){
      digitalWrite(12,HIGH);
      digitalWrite(13,HIGH);
      analogWrite(3,100);
      analogWrite(11,100);

}
void Left (){
      analogWrite(3,100);
      analogWrite(11,100);
      digitalWrite(12,HIGH);
      digitalWrite(13,LOW);

      }
```

Figure 1.26

```
void Backward (){
       digitalWrite(12,LOW);
       digitalWrite(13,LOW);
       analogWrite(3,100);
       analogWrite(11,100);
}

void Stop_robot (){
       analogWrite(3,0);
       analogWrite(11,0);
}

void Flaps_open()

{
 servo_9.write(180);
 servo_10.write(0);
}

void Flaps_closed()
{
servo_9.write(90);
servo_10.write(90);
}
```

Figure 1.27

Upload the final code for the **joystick** as shown in figure 1.28

```
// X axis reading
int VRX2 = 0;
// Y axis reading
int VRY2 =0;
// Switch reading
int SW2 =0;
void setup() {
  // put your setup code here, to run once:
  pinMode(13,OUTPUT);
  // Joystick 2
  pinMode(8,INPUT); // X axis pin
  pinMode(9,INPUT); // Y axis pin
  pinMode(4,INPUT); // switch pin
  Serial1.begin(115200);
  digitalWrite(4,HIGH);
}

void loop() {
  // put your main code here, to run repeatedly:
    // Reading the values
  VRX2 = analogRead(8);
  int   x_value2 = VRX2;
  VRY2 = analogRead(9);
  int  y_value2 = VRY2;
  SW2 = digitalRead(4);
  delay(10);

if(x_value2<100){
  Serial1.println('R');
  }
if(x_value2>800){
  Serial1.println('L');
}
if(y_value2<100){
  Serial1.println('B');
}
if(y_value2>800){
  Serial1.println('F');
}
if(x_value2>200&&x_value2<800&&y_value2>200&&y_value2<800){
  Serial1.println('S');
}

if(SW2 == 0){
  Serial1.println('O');
}
if(SW2 == 1){
  Serial1.println('C');
}
}
```


Figure 1.28

Workshop 8

Finally, we have finished working on building the soccerbot

Happy playing soccer Andy

Wow, I'm excited ! Time to kick some goals